

Deletion of the triplet repeat encoding polyglutamine within the mouse Huntington's disease gene results in subtle behavioral/motor phenotypes *in vivo* and elevated levels of ATP with cellular senescence *in vitro*

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Huntingtin (htt), the protein encoded by the Huntington's disease (HD) gene, contains a polymorphic stretch of glutamines (polyQ) near its N-terminus. When the polyQ stretch is expanded beyond 37Q, HD results. However, the role of the normal polyQ stretch in the function of htt is still unknown. To determine the contribution of the polyQ stretch to normal htt function, we have generated mice with a precise deletion of the short CAG triplet repeat encoding 7Q in the mouse HD gene (*Hdh*^{ΔQ}). *Hdh*(ΔQ/ΔQ) mice are born with normal Mendelian frequency and exhibit no gross phenotypic differences in comparison to control littermates, suggesting that the polyQ stretch is not essential for htt's functions during embryonic development. Adult mice, however, commit more errors initially in the Barnes circular maze learning and memory test and perform slightly better than wild-type controls in the accelerating rotarod test for motor coordination. To determine whether these phenotypes may reflect an altered cellular physiology in the *Hdh*^{ΔQ} mice, we characterized the growth and energy status of primary embryonic and adult *Hdh*(ΔQ/ΔQ) fibroblasts in culture. The *Hdh*^{ΔQ} fibroblasts exhibited elevated levels of ATP, but senesced prematurely in comparison with wild-type fibroblasts. Taken altogether, these results suggest that htt's polyQ stretch is required for modulating longevity in culture and support the hypothesis that the polyQ stretch may also modulate a htt function involved in regulating energy homeostasis.

INTRODUCTION

Huntington's disease (HD) is a fatal autosomal dominant neurodegenerative disorder characterized by midlife onset and a triad of cognitive, motor and emotional disturbances (1). HD is caused by the expansion of a CAG triplet repeat encoding polyglutamine (polyQ) within huntingtin (htt), the protein product of the *IT15* (HD) gene (2). Unaffected individuals have 6–34 CAG repeats while more than 37 repeats result in HD. HD is one of nine polyQ disorders that affect different neuronal populations in the central nervous system (3,4). Although the expanded stretch of polyQ in all of these disorders is thought to confer a deleterious gain-of-function, the

normal functions of these proteins likely play important roles in determining neuronal specificity in the different polyQ disorders.

Htt is a large (348 kDa), predominantly cytoplasmic protein that is also found at low steady-state levels in the nucleus (5,6). Htt is essential during early embryonic development (7–9) and is also required during development of the brain (10–12) and in spermatogenesis (10). Htt's normal function(s) may impact many cellular processes including signal transduction, endocytosis, cytoskeletal structure, transcription and axonal transport (13–15). The large number of putative functions for htt is also reflected in the large number of interacting proteins that have been characterized to date (16). Analysis of the htt

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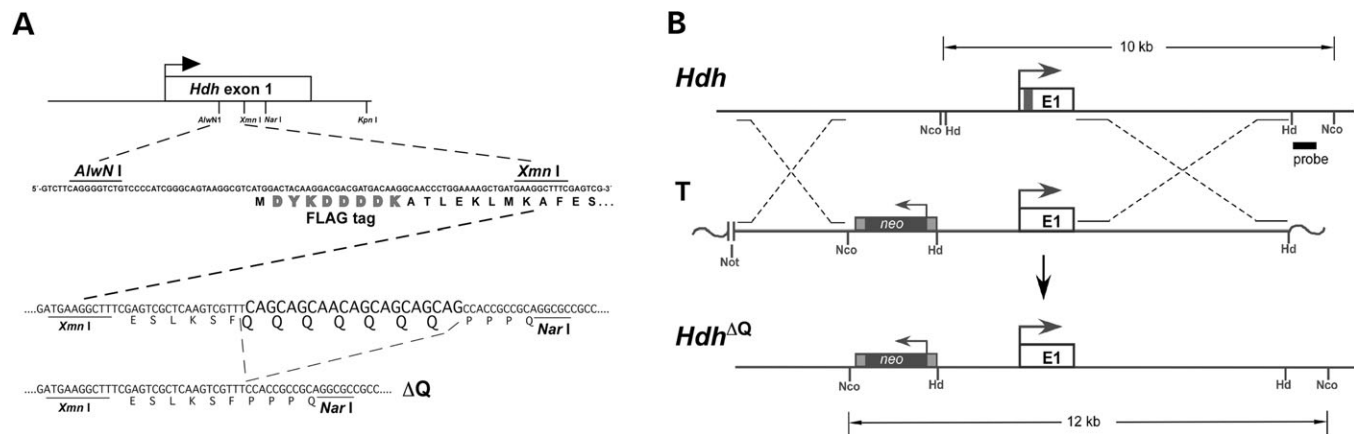


Figure 1. Strategy for deletion of the (CAG)₂CAA(CAG)₄ sequence encoding 7Q in the *Hdh* gene. (A) Diagram of the Δ Q and FLAG epitope tag sequence modifications introduced into *Hdh* exon 1. Key restriction sites used for the modifications are indicated. (B) Schematic of the wild-type *Hdh* allele near exon 1 (open box with gray region denoting the polyQ stretch) shown together with diagrams of the targeting construct (T) and the FLAG epitope- and Δ Q-modified *Hdh* allele (*Hdh* Δ Q). The position of the loxP-flanked (gray rectangles) *pgk-neo* selection cassette (*neo*) is indicated, along with the transcriptional orientations of *Hdh* and *neo* (arrows). The position of an *Hdh* 3'-flanking probe (black rectangle) and diagnostic restriction fragments identifying the wild-type and *Hdh* Δ Q alleles in Southern blot analyses are indicated. The targeting construct was linearized at a *NotI* restriction site (parallel bars) and the regions of 5'- and 3'-flanking homology of the targeting vector with *Hdh* genomic sequence are indicated with dashed crossed lines. Restriction enzyme sites are Nco, NcoI; Hd, *HindIII*; Not, *NotI*.

primary amino acid sequence reveals (sequentially from the N-terminus) a group of three lysine residues within the first 17 amino acids of htt that are a substrate for SUMOylation (17), a polyQ stretch followed closely by a proline-rich domain, a lipid-binding domain (18) located within one of 10 clumped HEAT repeats (an acronym based on members of a protein family containing similar ~ 38 amino acid repeats: Huntingtin, Elongation Factor 3A, A subunit of Protein Phosphatase 2A and TOR1) (19) and a nuclear export signal near the C-terminus (5). The HEAT repeats favor an alpha helical structure and are implicated in modulating protein–protein interactions (20). This has led to the hypothesis that htt, like other HEAT domain containing proteins, can act as a scaffold mediating various cellular processes. The role of any of these domains in the normal function of htt has yet to be determined *in vivo*.

The polyQ stretch is conserved among vertebrates, ranging from an average length of approximately 20 glutamines in humans to only four in zebrafish and pufferfish (16,21,22). Interestingly, *Drosophila* htt does not have a polyQ stretch (23), suggesting the possibility that the polyQ stretch may have a function unique to vertebrates. In lymphoblastoid cell lines derived from HD patients, the length of the polyQ stretch influences cellular energy status, with an inverse relationship between polyQ length and the cell's ATP/ADP ratio. Moreover, this relationship is also valid for short stretches of polyQ (<35Q), suggesting that polyQ length can modulate energy status even in normal cells (24). The impact of deleting polyQ on the normal function of a vertebrate htt, however, has not yet been explored.

To begin investigating the contribution of the polyQ stretch to normal htt function *in vivo*, we have generated mice with a precise deletion of the (CAG)₂CAA(CAG)₄ triplet repeat encoding 7Q within the mouse homolog of the HD gene (*Hdh* Δ Q). We have found that the polyQ stretch is not essential for normal htt function during development, but *Hdh*(Δ Q/ Δ Q) homozygous adult mice exhibit more errors initially in a

learning/memory test and show a subtly enhanced motor coordination phenotype in the accelerating rotating rod (rotarod) test. In addition, primary fibroblasts obtained from embryos homozygous for the Δ Q deletion exhibit an elevated level of ATP and senesce more rapidly than wild-type controls. Our results support the hypothesis that the polyQ stretch within htt is required for modulating proliferative life span in primary cell culture and that one of htt's normal functions may also be involved in regulating energy homeostasis in vertebrates.

RESULTS

Derivation of mice carrying the *Hdh* Δ Q allele

To generate a precise deletion of the *Hdh* (CAG)₂CAA(CAG)₄ triplet repeat encoding 7Q, a gene-targeting vector was assembled, which contained both the deletion of the (CAG)₂CAA(CAG)₄ repeat (Δ Q) and, for the purpose of following expression of Δ Q-htt, a FLAG epitope tag located at the N-terminus of the coding sequence within exon 1 (Fig. 1A). We note that the addition of an N-terminal FLAG tag has the potential to alter normal htt localization *in vivo*. However, in cell culture, a FLAG tag at the N-terminus of htt does not interfere with its expression or localization (18,25). Moreover, mice carrying a triple FLAG tag (FLAG3X) at the N-terminus of endogenous htt appear phenotypically normal (manuscript in preparation). For positive selection of transfected ES cells, a neomycin phosphotransferase gene cassette flanked by loxP sites (floxed *pgkneo*) was inserted ~ 1.3 kb upstream of the *Hdh* transcriptional start site and in the opposite transcriptional orientation with respect to *Hdh* (Fig. 1B). Although the loxP sites provide a means to remove the *pgkneo* cassette via Cre-mediated recombination, the location of the *pgkneo* cassette does not interfere with *Hdh* expression (26,27) and provides a convenient

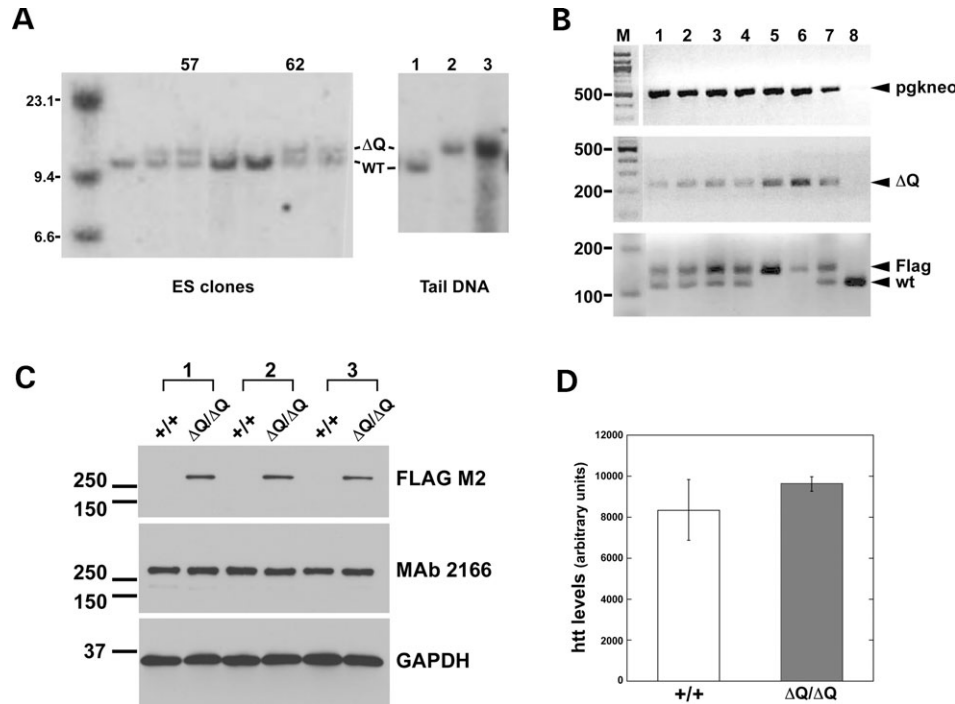


Figure 2. Genotyping and western analyses of *Hdh*^{ΔQ} ES clones and mouse tissue. (A) Southern blot of ES clones (left) and tail DNA prepared from selected progeny of an *Hdh*(ΔQ/+) intercross (right). DNA from G418-resistant ES colonies and tail biopsies was digested with *Nco*I and hybridized with a 3'-flanking probe that recognizes the 10 kb wild-type (WT) and 12 kb *Hdh*^{ΔQ} genomic fragments (ΔQ). The two ES clones used to obtain germline transmission of the *Hdh*^{ΔQ} allele (clones 57 and 62) are indicated, along with the sizes of ³²P-labeled *Hind*III restriction fragments of λ-phage DNA (left). On the right, lane 1 represents an *Hdh*(+/+) mouse, whereas lanes 2 and 3 are *Hdh*(ΔQ/ΔQ). (B) PCR analyses of eight tail biopsies from a litter obtained from an *Hdh*(ΔQ/+) intercross using oligonucleotide primers amplifying a region of the *pgk-neo* cassette (pgkneo), the polyQ deletion (ΔQ) and across the region of exon 1 where the FLAG epitope tag was inserted (Flag and wt products). Lanes 1–4 and 7: *Hdh*(ΔQ/+) pups, lane 8: wild-type pup and lanes 6 and 7: *Hdh*(ΔQ/ΔQ) homozygotes. The sizes (in base pairs) of marker DNA fragments (lane M) are indicated on the left. (C) Western analysis of cytoplasmic protein extracts from whole brains obtained from wild-type, (n = 3, +/+) and *Hdh*(ΔQ/ΔQ), (n = 3, ΔQ/ΔQ) mice. Fifty micrograms of each extract was fractionated on 9% SDS–PAGE, blotted to PVDF membranes and then strips of membrane were probed with antibodies recognizing ΔQ-htt (top strip, Mab FLAG M2), both wild-type and ΔQ-htt (middle strip, Mab 2166) and GAPDH (bottom strip, Mab374). The sizes (in kDa) of protein standards are indicated on the left. (D) Histogram of htt levels in wild-type (+/+) and *Hdh*(ΔQ/ΔQ) whole brain extracts calculated from the blots in (C). Htt levels are expressed in arbitrary units obtained from scanning film exposures of the western blots. There was no significant difference in wild-type and ΔQ-htt levels (*P* = 0.27). Error bars represent SD.

marker for routine genotyping. Targeted ES cell clones were identified by Southern and PCR analyses (Fig. 2A and B). For PCR detection of the ΔQ mutation, an oligonucleotide primer spanning the site of the (CAG)₂CAA(CAG)₄ repeat deletion was used to amplify specifically sequence from the mutant allele. In addition, primers flanking the epitope tag insertion site were used to discriminate between the wild-type and *Hdh*^{ΔQ} alleles. Germline transmission of the *Hdh*^{ΔQ} allele (*Hdh*^{tm3Szi}, GenBank accession no. MGI: 3603606) was obtained from two targeted ES cell lines. All mice analyzed in this study were of a mixed strain background (C57BL6/129Sv).

The polyQ domain is not essential for htt function during embryonic development

Htt is essential for embryonic development, and mice lacking htt expression die between embryonic day 7.5 (E7.5) and E9, shortly after the onset of gastrulation (7–9). If htt's embryonic function is compromised by the ΔQ mutation, the fraction of *Hdh*(ΔQ/ΔQ) homozygotes obtained from an *Hdh*(ΔQ/+) intercross should be lower than the predicted Mendelian

distribution of 25%, and any surviving *Hdh*(ΔQ/ΔQ) neonatal pups may exhibit phenotypic abnormalities. In an *Hdh*(ΔQ/+) intercross, approximately one-quarter of the progeny were *Hdh*(ΔQ/ΔQ) (Table 1), and moreover, *Hdh*(ΔQ/ΔQ) pups were also indistinguishable from *Hdh*(ΔQ/+) and wild-type littermates at birth (data not shown). These results indicate that lack of the polyQ stretch does not interfere substantially with htt's normal function during development. Genotypes were obtained by both Southern and by PCR analyses to confirm the presence of the FLAG tag, the *pgkneo* cassette and the ΔQ mutation (Fig. 2A and B). To ensure that the ΔQ mutation and FLAG tag did not affect ΔQ-htt expression, western analysis of whole brain extracts from *Hdh*(ΔQ/ΔQ) and age-matched wild-type littermate controls (n = 3) was performed. An anti-FLAG antibody recognized a htt-sized polypeptide only in the *Hdh*(ΔQ/ΔQ) brain extracts, whereas an anti-htt antibody (Mab 2166) reacts with both wild-type and ΔQ-htt in both the *Hdh*(+/+) and *Hdh*(ΔQ/ΔQ) samples (Fig. 2C). Densitometry of the western exposures, using the level of glyceraldehyde phosphate dehydrogenase (GAPDH) as a loading control, revealed that there was no significant difference in the amounts of htt in the samples (*P* = 0.27,

Table 1. Genotyped progeny

Cross	No. of litters	No. of genotyped ^a (No. of expected ^b) progeny		
		<i>Hdh</i> (+/+)	<i>Hdh</i> (ΔQ/+)	<i>Hdh</i> (ΔQ/ΔQ)
<i>Hdh</i> (ΔQ/+) X <i>Hdh</i> (ΔQ/+)	25	39 (32)	57 (65)	33 (32)

^aAll progeny genotyped at postnatal day 10.

^bCalculated from the predicted Mendelian ratio. Observed number of progeny is consistent with the predicted Mendelian ratio (χ^2 $P < 0.05$ for all genotypes).

Fig. 2D), thus indicating that the polyQ deletion does not affect htt expression levels.

The ΔQ mutation does not affect htt’s subcellular localization

Wild-type htt is a predominantly cytoplasmic protein that is localized in the perinuclear region. To determine whether the polyQ deletion affected the subcellular localization of ΔQ-htt, primary fibroblasts were prepared from embryonic day 13 (E13) *Hdh*(ΔQ/ΔQ) and wild-type embryos. Subcellular fractionation of passage 4 (P4) cells revealed no difference in the cellular compartmentalization of wild-type and ΔQ-htt, with most htt residing in the cytoplasm (Fig. 3A). Grb2, an SH2/SH3 domain adaptor protein (28), and GAPDH were used as a positive controls for predominantly cytoplasmic proteins, whereas p53 is predominantly nuclear (29), and htt interacting protein-1 (HIP-1) is an example of a protein that partitions both the cytoplasm and the nucleus by virtue of a nuclear localization signal at its C-terminus (30). In our primary mouse embryonic fibroblasts (PMEFs), the majority of HIP-1 is found in the nucleus. To confirm that wild-type htt and ΔQ-htt had similar subcellular localization properties, immunohistochemical analyses were performed on *Hdh*(ΔQ/ΔQ) and *Hdh*(+/+) fibroblasts and brain sections using MAb 2166. Htt staining was punctate and perinuclear in both *Hdh*(ΔQ/ΔQ) and control P4 fibroblasts, and predominantly cytoplasmic in neurons, confirming that there is no significant difference in the subcellular localization of wild-type and ΔQ-htt (Fig. 3B).

Although ΔQ-htt’s subcellular localization is indistinguishable from wild-type htt, deletion of the polyQ stretch may affect ΔQ-htt’s associations with interacting proteins, especially those known to be sensitive to the length of the polyQ stretch. To examine ΔQ-htt’s interactions with protein partners, we performed co-immunoprecipitation analyses with wild-type and *Hdh*(ΔQ/ΔQ) whole brain extracts using MAb 2166. Both wild-type and ΔQ-htt were immunoprecipitated with similar efficiency, and there were no significant differences observed in the co-precipitation of HIP-1, PSD-95 and GAPDH (Fig. 3C). Although neither HIP-1 nor PSD-95 interacts directly with polyQ, the strength of their interaction with htt is reduced when htt’s polyQ stretch is expanded into the disease range (31,32). GAPDH interaction with ΔQ-htt was examined on the basis of previous reports suggesting that GAPDH associates with htt via the polyQ stretch (33,34). However, our results suggest that other regions within htt must be involved, as the interaction of GAPDH with ΔQ-htt is comparable to its interaction with

wild-type htt. Thus, although expansion of the polyQ stretch within htt is known to perturb htt’s binding to HIP-1, PSD-95 and GAPDH, deletion of the polyQ stretch does not have a similar effect on htt’s interaction with these proteins.

***Hdh* (ΔQ/ΔQ) mice exhibit subtle behavioral and motor phenotypes**

There was no significant difference in weights found between *Hdh*(ΔQ/ΔQ) females and wild-type females or between *Hdh*(ΔQ/ΔQ) males and wild-type males at any of the ages that were examined (Table 2). Limb claspings, a behavior indicative of neurological dysfunction in rodents, which is seen often in expanded polyQ and HD mouse models (35–37), including a conditional htt loss-of-function model (10), was not observed in *Hdh*(ΔQ/ΔQ) mice up to the oldest age examined (1 year of age). Gross motor deficits were not observed in *Hdh*(ΔQ/ΔQ) mice subjected to a cage-top rotation test and a wire-rod hanging test (38). In addition, no obvious anatomical abnormalities were detected in Cresyl violet-stained sections of brains obtained from 1-year-old *Hdh*(ΔQ/ΔQ) mice and wild-type littermate controls (Fig. 4). Moreover, there was no significant difference in brain weights obtained from female wild-type (+/+) and *Hdh*(ΔQ/ΔQ) littermates (+/+ = 0.43 ± 0.03 g; ΔQ/ΔQ = 0.45 ± 0.08 g; $P = 0.70$ for brain weight and $P = 0.59$ for brain weight as a percentage of body weight; $n = 3$ for each genotype).

To determine whether a more subtle phenotype could be detected by behavioral and cognitive testing, *Hdh*(ΔQ/+), *Hdh*(ΔQ/ΔQ) and wild-type control mice were subjected to tests measuring overall anxiety (elevated plus maze, light/dark box, open field and activity cage), a test of spatial learning and memory (Barnes circular maze), tests for olfaction and a test of motor coordination (accelerating rotarod) (38).

Anxiety-like behaviors were measured at 3, 6 and 9 months of age. These tests take advantage of the tension between the natural tendency the mice have to explore a novel environment and their innate preference for the dark and avoidance of open spaces (38). There were no significant differences found between *Hdh*(ΔQ/ΔQ) mice and wild-type controls at 3 months of age in the elevated plus maze based on the amount of time spent in the open arms ($P = 0.88$) or the total number of arm entries ($P = 0.67$). There were also no differences found at 6 months of age in the light–dark box in the latency to enter the dark box ($P = 0.99$), the amount of time spent in the dark box ($P = 0.66$) or the number of exits from the dark box ($P = 0.60$). Testing in the open field also revealed no significant differences between deletion

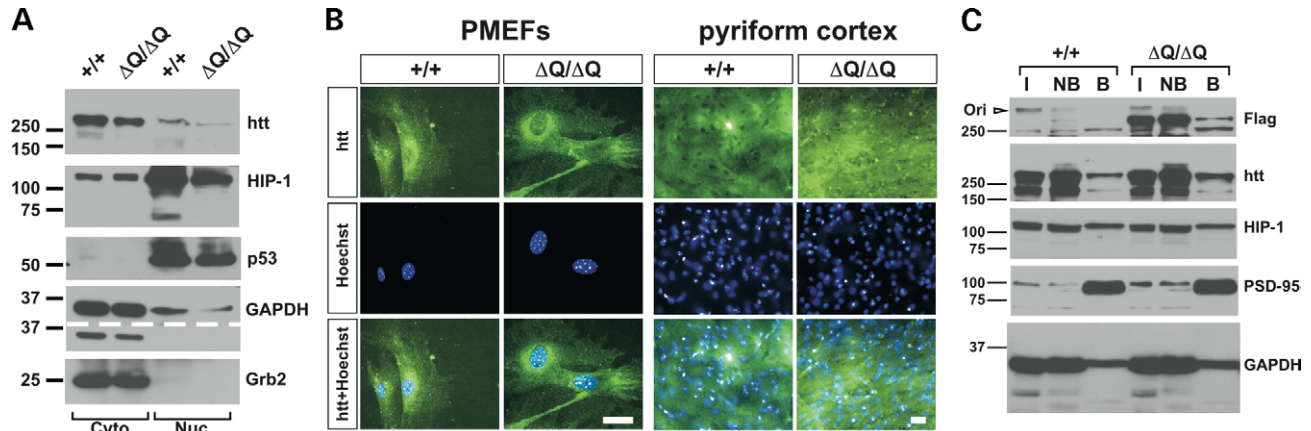


Figure 3. Deletion of htt's polyQ stretch does not affect its subcellular localization or association with htt-interacting proteins. **(A)** Cytoplasmic (Cyto) and nuclear (Nuc) protein extracts (~50 μg) from wild-type, (+/+) and *Hdh*(ΔQ/ΔQ), (ΔQ/ΔQ) PMEFs were fractionated on 4–20% SDS-PAGE and transferred to a PVDF membrane. Portions of the membrane were then probed with an antibody to htt (MAb 2166, top strip) and antibodies to proteins that associate preferentially with the nuclear fraction (HIP-1, p53) and antibodies to proteins that are predominantly cytoplasmic (GAPDH, Grb2). Both a longer and a shorter exposure of the membrane strip (separated by a white dotted line) probed with the GAPDH antibody is shown to indicate that slightly less protein was loaded in the ΔQ/ΔQ sample lanes. The sizes (in kDa) of protein standards are indicated on the left. **(B)** PMEFs (left panels) obtained from wild-type and *Hdh*(ΔQ/ΔQ) embryos were probed with an anti-htt antibody (MAb 2166, green) and stained with Hoechst to visualize nuclei (blue). Sections displaying different regions of the pyriform cortex from wild-type and *Hdh*(ΔQ/ΔQ) brains (right panels) were probed with MAb 2166 (green) and stained with Hoechst (blue). Scale bars = 50 μm. **(C)** Co-immunoprecipitation of htt-interacting proteins with wild-type htt or ΔQ-htt in *Hdh*(+/+) and *Hdh*(ΔQ/ΔQ) whole brain extracts (9-month-old mice), respectively, using an anti-htt antibody recognizing both wild-type and ΔQ-htt (MAb 2166). Fifty micrograms of input (I) protein (representing 5% of the input), 50 μg of antibody non-bound fraction (NB, representing 5% of this fraction) and antibody-bound protein (B, representing 35% of the immunoprecipitated protein) were fractionated on a 9% SDS-PAGE gel and then transferred to PVDF membrane. Membrane strips were probed with an anti-FLAG antibody (top strip, Flag), MAb 2166 antibody (htt) and antibodies to HIP-1, PSD-95 and GAPDH. The sizes (in kDa) of protein standards are indicated on the left, along with the origin of the separating gel (Ori). Data are representative of three independent experiments.

Table 2. Weight (g) of wild-type and *Hdh*(ΔQ/ΔQ) littermates

Sex	Genotype (n)	Weight (SE)			
		1 month	3 months	6 months	9 months
F	+/+ (11)	15.2 (0.6)	22.6 (0.8)	24.9 (1.0)	25.5 (1.5)
F	ΔQ/ΔQ (8)	15.5 (0.9)	20.3 (0.8)	23.1 (0.7)	23.6 (0.8)
M	+/+ (6)	18.9 (1.5)	27.1 (1.5)	31.8 (2.2)	31.2 (1.2)
M	ΔQ/ΔQ (11)	17.1 (0.5)	26.8 (0.8)	30.2 (1.1)	29.9 (1.7)

mutants and controls at 9 months of age based on the amount of time spent in the center squares ($P = 0.86$). Similarly, there were no significant differences in overall horizontal and vertical activity between wild-type littermates and *Hdh*(ΔQ/ΔQ) mice when measured in an automated activity cage at 9 months of age ($P = 0.46$).

Cognitive abilities were assessed using the Barnes circular maze, a measure of spatial learning and memory (39) (Fig. 5A). To avoid a bright light and buzzing sound in this test, mice must find the correct hole in a circular platform leading to a hidden escape tunnel. No significant differences in learning and memory were found at 5 months of age based on a distance score from the target (two-way RM ANOVA for genotype, $P = 0.16$; Fig. 5B) (see Materials and Methods). The distance score relies heavily on spatial cues and proper hippocampal function and appears to be unimpaired. However, when the number of errors was examined, it was apparent that the *Hdh*(ΔQ/+) and *Hdh*(ΔQ/ΔQ) mutants would explore many more of the holes in the maze before entering the escape tunnel on the first 2 trial days (Fig. 5C). Besides the significant difference found in the analysis of

trial day, which indicates performance improved over the course of the test for all groups [$F_{(8,5)} = 25.02$; $P < 0.001$], an ANOVA revealed a group difference in genotype [$F_{(2,5)} = 21.1$; $P < 0.001$], and a significant interaction was found between trial day and genotype [$F_{(8,5)} = 4.04$; $P < 0.001$]. This indicates that the *Hdh*(ΔQ/+) and *Hdh*(ΔQ/ΔQ) mice performed poorly compared with wild-type controls, as an increase in the number of errors is considered a deficit in this type of learning and memory test, even in the absence of a distance score deficit. It is unlikely that this behavior was due to the decreased anxiety because the *Hdh*(ΔQ/ΔQ) mice behaved similarly to the controls in the elevated maze, light–dark box and open-field tests. Interestingly, the number of errors generated by *Hdh*(ΔQ) mice and controls were comparable in subsequent days of testing (Fig. 5C).

A similar tendency for increased exploration of a novel environment by *Hdh*(ΔQ/ΔQ) mice was observed in an olfaction test (38). Olfaction was assessed by a paradigm requiring mice to find food buried in the cage bedding at 7 months of age ($n = 5$) after 24 h of food deprivation. The *Hdh*(ΔQ/ΔQ) mice were slow to find a chunk of Nutri-grain bar, tending to explore the cage far more than the wild-type controls ($+/+ = 38.4 \pm 9.5$ s; $\Delta Q/\Delta Q = 143 \pm 20.7$ s; $P = 0.004$). In some instances, the *Hdh*(ΔQ/ΔQ) mice did not eat the treat, although the controls always did. There was no difference, however, in their latency to dig in the clean bedding ($+/+ = 23.6 \pm 4.7$ s; $\Delta Q/\Delta Q = 33.2 \pm 10.3$ s; $P = 0.43$). The test was repeated at 8 months of age using a food deprivation period of 36 h and a more palatable treat (chocolate) to ensure that the animals were properly motivated to find the hidden food. As in the previous test, the latency to

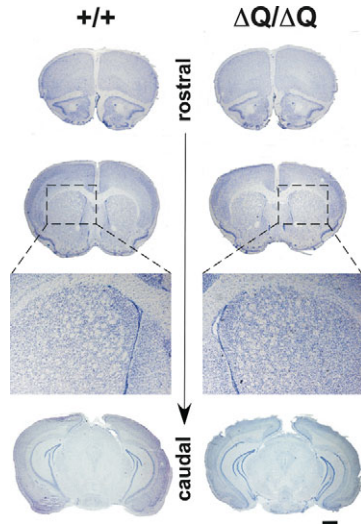


Figure 4. Normal brain morphology in *Hdh*($\Delta Q/\Delta Q$) mice. Cresyl violet-stained rostral and caudal sections from wild-type, (+/+) and *Hdh*($\Delta Q/\Delta Q$) mouse brains. For comparison, a region of the striatum (indicated with dashed rectangle) is shown at higher magnification. Scale bar = 1 mm.

dig in the bedding did not differ significantly between genotypes (+/+ = 13.2 ± 3.1 s; $\Delta Q/\Delta Q$ = 24.4 ± 6.5 s; $P = 0.17$), and although not reaching significance, there was a trend toward the *Hdh*($\Delta Q/\Delta Q$) mice being slower to find the treat and eat it (+/+ = 21.2 ± 5 s; $\Delta Q/\Delta Q$ = 68.8 ± 29.2 s; $P = 0.07$).

To determine whether the increased latency to find a food treat exhibited by the *Hdh*($\Delta Q/\Delta Q$) mice reflected a true olfactory deficit or a motivational/exploratory anomaly, olfaction was also assessed using a test that did not require the mice to locate a reward ($n = 5$, mice from the same cohort used in the buried treat test). The mice were placed in a wire cage with clean bedding covering one-half of the cage and their own bedding covering the other half. Typically, mice will prefer to remain on the side with their own bedding, and in this test, there was no difference in the amount of time the *Hdh*($\Delta Q/\Delta Q$) mice or wild-type controls spent on top of their own bedding (+/+ = 97 ± 15 s; $\Delta Q/\Delta Q$ = 110 ± 7.4 s; $P = 0.23$). This result suggests that the *Hdh*($\Delta Q/\Delta Q$) mice have no significant deficit in olfaction and their increased latency to find a treat may represent a difference in motivational/exploratory behavior.

To assess motor coordination in the *Hdh*($\Delta Q/\Delta Q$) mice, they were tested at 1, 4 and 9 months of age on an accelerating rotarod. The rotarod test assesses motor coordination and balance, as well as motor learning (38). Surprisingly, at some time points, the *Hdh*($\Delta Q/\Delta Q$) mice performed slightly better on this test than their wild-type littermate controls (Fig. 5D). At 1 and 9 months of age, an overall ANOVA showed differences in rotarod performance. These differences were not reflected in the 4 months data (data not shown).

Both *Hdh*($\Delta Q/\Delta Q$) and wild-type mice showed an increase in the latency to fall in succeeding trial days at each age examined, indicating that both groups of mice were learning how to stay on the rod longer. Analysis by ANOVA showed a significant effect of trial day at 1 month [$F_{(4,16)} = 96.9$;

$P < 0.001$], 4 months [$F_{(4,14)} = 56.6$; $P < 0.001$] and 9 months [$F_{(4,12)} = 78.22$; $P < 0.001$].

The *Hdh*($\Delta Q/\Delta Q$) mice performed better on the rotarod at 1 and 9 months of age, but curiously not at 4 months of age. There was a significant effect of genotype at 1 month of age [$F_{(1,16)} = 8.9$; $P = 0.009$] and at 9 months of age [$F_{(1,12)} = 9.6$; $P = 0.009$], but not at 4 months of age [$F_{(1,14)} = 0.09$; $P = 0.76$]. Specifically, the *Hdh*($\Delta Q/\Delta Q$) mice did better on trial days 1 ($P = 0.039$) and 2 ($P = 0.05$) at 1 month and on trial day 3 ($P = 0.03$) at 9 months. There was also a significant genotype versus trial day interaction at 1 month of age [$F_{(4,16)} = 7.3$; $P < 0.001$], 4 months of age [$F_{(4,14)} = 3.3$; $P = 0.017$] and at 9 months of age [$F_{(4,12)} = 2.8$; $P = 0.037$].

Hdh($\Delta Q/\Delta Q$) fibroblasts have elevated levels of ATP

The behavioral and motor phenotypes exhibited by the *Hdh*($\Delta Q/\Delta Q$) mice may reflect specific changes in CNS function or a more global change in cellular metabolism. To begin to address the basis for these phenotypes, we examined the basic metabolic status of *Hdh*($\Delta Q/\Delta Q$) primary embryonic fibroblasts by measuring ATP levels in *Hdh*($\Delta Q/\Delta Q$) and wild-type control cell cultures. In initial experiments with wild-type and *Hdh*($\Delta Q/\Delta Q$) fibroblasts aimed at comparing the subcellular localization of htt and ΔQ -htt, we noticed that the *Hdh*($\Delta Q/\Delta Q$) fibroblasts were less dense in culture (37% the density of the wild-type cultures) and had an altered morphology at later passages in culture. We measured ATP levels in the cells using a bioluminescent read-out assay at P7 and found that the *Hdh*($\Delta Q/\Delta Q$) fibroblasts had a significantly increased level of ATP/cell (wild-type = $6.4 \times 10^{-11} \pm 6.3 \times 10^{-12}$ moles ATP/cell and $\Delta Q/\Delta Q$ = $9.16 \times 10^{-11} \pm 5.3 \times 10^{-12}$ moles ATP/cell; $P = 0.007$). To determine whether the change in ATP levels also occurred at earlier passages, we measured ATP levels in wild-type, *Hdh*($\Delta Q/+$) and *Hdh*($\Delta Q/\Delta Q$) embryonic fibroblasts obtained from *Hdh*($\Delta Q/+$) intercrosses (embryos from two females, $n = 3$ embryos for each genotype). In early passage fibroblasts (P2–P4), ATP levels were similar in all genotypes ($P = 0.17$ – 0.41) (Fig. 6A). At P5, however, the ATP level in the *Hdh*($\Delta Q/\Delta Q$) fibroblasts was increased relative to the levels observed in the *Hdh*($\Delta Q/+$) and wild-type control cells ($P = 0.01$), and at P6, the level of ATP in the *Hdh*($\Delta Q/\Delta Q$) cells was ~4-fold higher than the level in the *Hdh*($\Delta Q/+$) and wild-type cells ($P = 0.0006$) (Fig. 6A).

To determine whether primary cultures of cells derived from adult mice also have increased levels of ATP, fibroblasts from ear biopsies (13 months old, $n = 1$ from each genotype) were obtained and ATP levels were assessed. In P1 cultures, ATP levels in the adult *Hdh*($\Delta Q/\Delta Q$) fibroblasts were again elevated ~4-fold over wild-type ATP levels ($P < 0.0001$) (Fig. 6B). However, in contrast to the results with primary cells in culture, ATP levels in cortex and skeletal muscle from 14-month-old wild-type and *Hdh*($\Delta Q/\Delta Q$) mice were not significantly different ($P = 0.86$ and $P = 0.78$, respectively) (Fig. 6C). Therefore, the increase in ATP levels exhibited in primary embryonic and adult fibroblasts may reflect a feature that is elicited by the stress of *in vitro* cell culture.

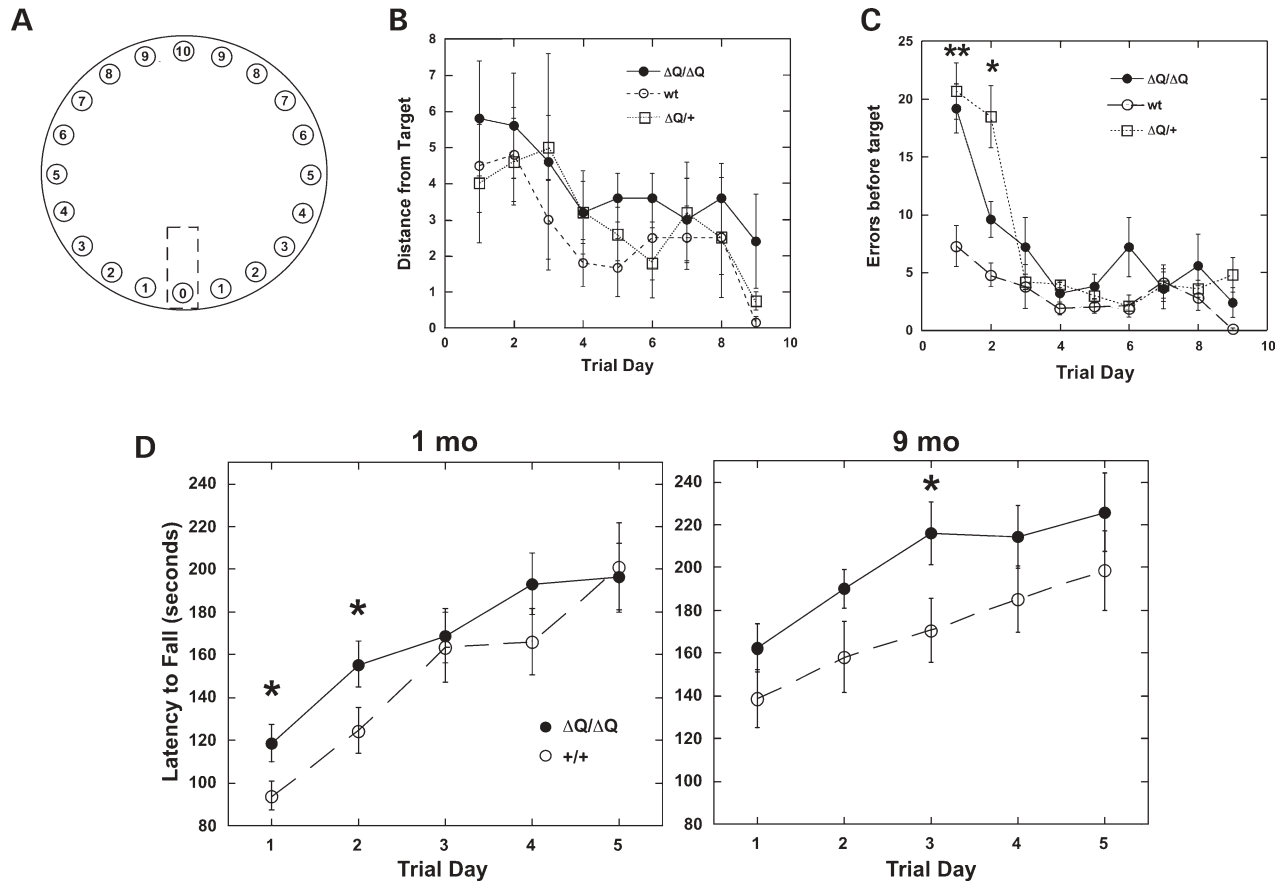


Figure 5. *Hdh*($\Delta Q/\Delta Q$) mice exhibit subtle behavioral and motor phenotypes. (A) Diagram of the Barnes circular maze showing the 20 escape holes along the periphery, one of which leads to an escape tunnel (dotted line). The numbers represent distance scores assigned to each escape hole, with the largest numerical value assigned to the hole that is located furthest along the periphery from the escape tunnel. (B) Plot of the distance scores for wild-type (wt), *Hdh*($\Delta Q/+$) and *Hdh*($\Delta Q/\Delta Q$) mice in the Barnes circular maze test. Although the *Hdh*($\Delta Q/\Delta Q$) mice tended to have higher distance scores compared with controls, the difference between control and *Hdh*($\Delta Q/\Delta Q$) distance scores over the nine trials was not significant ($n = 6$; $P = 0.162$). (C) Plot of the number of errors produced by wild-type (wt), *Hdh*($\Delta Q/+$) and *Hdh*($\Delta Q/\Delta Q$) mice in the Barnes circular maze. Both *Hdh*($\Delta Q/+$) and *Hdh*($\Delta Q/\Delta Q$) mice produced more errors in the first two trials when daily performance was examined using ANOVA ($n = 6$; $P = 0.001$). Significant differences between the ΔQ -mice and controls on individual trial days are indicated with asterisks (** $P = 0.001$, * $P < 0.05$; Student's *t*-test). (D) Plots of the latency to fall from an accelerating rotarod for wild-type control and *Hdh*($\Delta Q/\Delta Q$) mice at 1 month ($n = 17$) and 9 months ($n = 13$) of age. At 1 and 9 months, the *Hdh*($\Delta Q/\Delta Q$) mice exhibited slightly enhanced performance relative to their wild-type littermates (ANOVA, see text; $P < 0.009$). Significantly enhanced performance on individual trial days is indicated with an asterisk (* $P \leq 0.05$, Student's *t*-test). Error bars in (B–D) represent the SE.

Hdh($\Delta Q/\Delta Q$) fibroblasts senesce prematurely

The increase in cellular ATP levels in the *Hdh*($\Delta Q/\Delta Q$) fibroblasts was accompanied by a decrease in cell proliferation. Although both *Hdh*($\Delta Q/+$) and *Hdh*($\Delta Q/\Delta Q$) primary embryonic fibroblasts grew more slowly than wild-type cells at P2 ($P = 0.02$ and 0.04 , respectively), *Hdh*($\Delta Q/+$) cell growth was not significantly different from wild-type cells at later passages ($P = 0.36$ – 1.0 , Fig. 6D). In contrast, *Hdh*($\Delta Q/\Delta Q$) embryonic fibroblasts continued to divide slowly after P2, and at P6, *Hdh*($\Delta Q/\Delta Q$) cells were $\sim 30\%$ the density of the wild-type cultures ($P = 0.025$, Fig. 6D). In addition, many of the *Hdh*($\Delta Q/\Delta Q$) cells at P4–P6 exhibited an altered morphology and were observed to have double or multiple nuclei (Fig. 7A).

To determine whether these phenotypes were related to cellular senescence, *Hdh*($\Delta Q/\Delta Q$) and control fibroblast cultures were stained for senescence-associated β -galactosidase

(SA- β gal) activity (Fig. 7B). Although the role of SA- β gal expression in cellular senescence is unclear, the presence of SA- β gal activity correlates well with cellular aging in humans and mice (40). In senescent cells, the fraction of cells staining robustly for pH 6.0 SA- β gal activity was 2.5-fold higher than in the wild-type control culture [*Hdh*(+/+) = $14 \pm 2.9\%$ and *Hdh*($\Delta Q/\Delta Q$) = $37 \pm 3.9\%$ cells positive for SA- β gal staining; $P < 0.001$].

In addition to increased SA- β gal expression, elevated levels of p21^{Waf1/Cip1} (41) and p16^{INK4A} (42) are often observed in senescent cells. p21 is a cyclin-dependent kinase inhibitor that interacts with cdk2-associated complexes and prevents the cell cycle from entering into S phase. Depending upon the tissue and cell type, p21 is also involved in differentiation and apoptosis (43). Immunofluorescence staining for p21 expression in wild-type and *Hdh*($\Delta Q/\Delta Q$) fibroblasts at P6 revealed an increase in nuclear staining for p21 in the mutant fibroblast cultures (Fig. 7C). p21 is also a target gene of p53

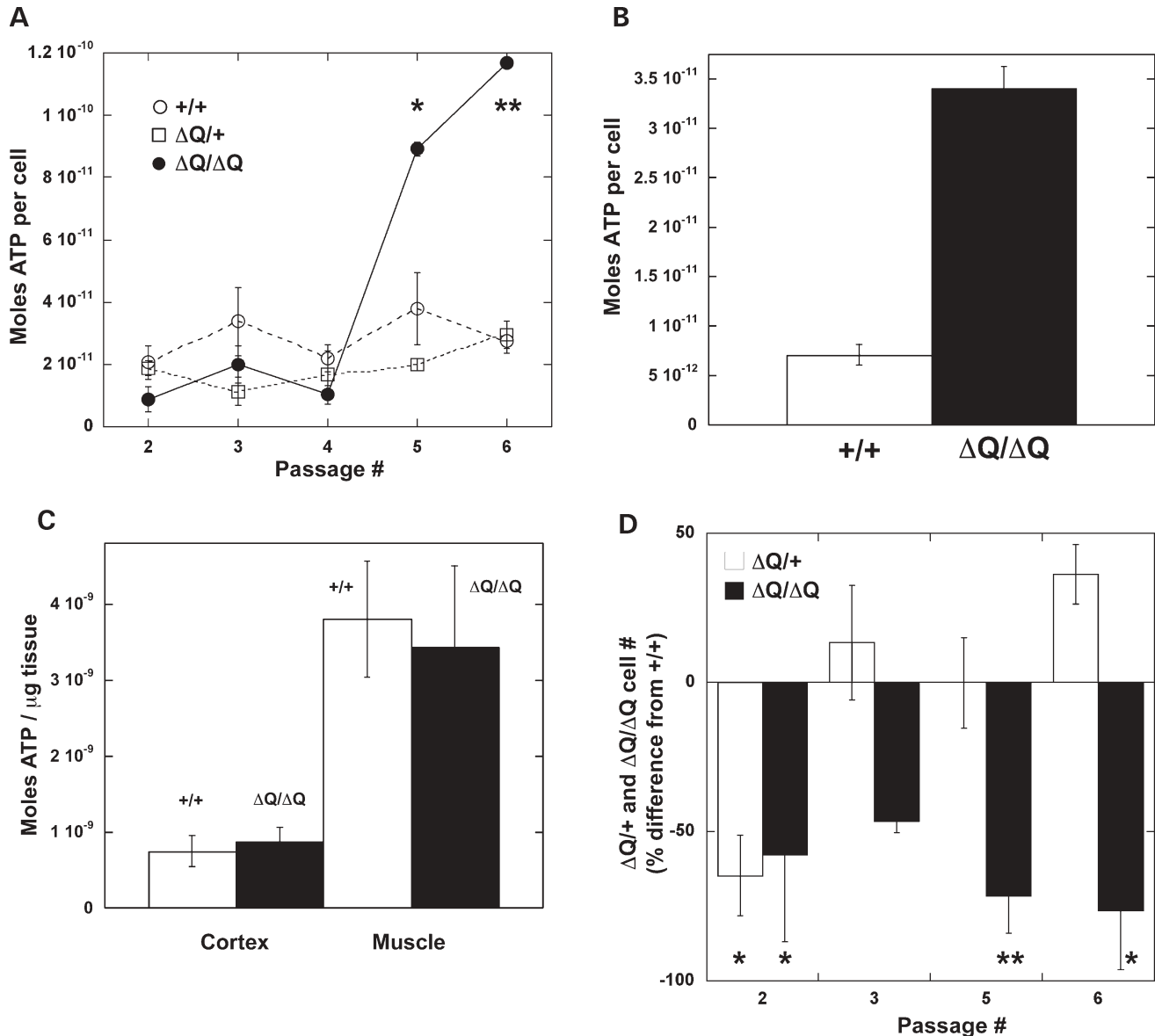


Figure 6. Increased ATP levels and reduced proliferation in *Hdh*($\Delta Q/\Delta Q$) fibroblasts. (A) Plot of ATP levels in cell extracts obtained from *Hdh*($\Delta Q/\Delta Q$) ($n = 3$, closed circles), *Hdh*($\Delta Q/+$) ($n = 3$, open squares) and wild-type cells ($n = 3$, open circles) versus passage number. Values are mean $\pm$ SEM in quadruplicate assays. Significantly increased levels of ATP were observed in *Hdh*($\Delta Q/\Delta Q$) PMEFs at P5 and P6 (* $P < 0.05$, ** $P < 0.007$; one-way ANOVA). (B) Histogram showing ATP levels in adult ear fibroblasts obtained from a 13-month-old wild-type (open bar) and *Hdh*($\Delta Q/\Delta Q$) mouse (black bar). Values are mean $\pm$ SEM in quadruplicate assays. (C) Histogram showing ATP levels in frontal cortex (Cortex) and skeletal muscle (Muscle) tissue obtained from 14-month-old wild-type (+/+) ($n = 4$) and *Hdh*($\Delta Q/\Delta Q$) mice ($n = 5$). Values are mean $\pm$ SEM in quadruplicate assays. (D) Histogram showing the difference between *Hdh*($\Delta Q/\Delta Q$) (black bars), *Hdh*($\Delta Q/+$) (white bars) and wild-type PMEF cell number (expressed as a percentage change from wild-type cell number) at various passage numbers. Both *Hdh*($\Delta Q/+$) and *Hdh*($\Delta Q/\Delta Q$) cell number was significantly less than wild-type at P2, whereas only *Hdh*($\Delta Q/\Delta Q$) cell number was reduced significantly relative to wild-type at P5 and P6 (* $P < 0.04$, ** $P < 0.01$; ANOVA). Cell numbers were obtained from the same cultures used to determine ATP levels in (A).

and can mediate p53's function as a tumor suppressor (44). The relative levels of p53, however, were not obviously different in wild-type and *Hdh*($\Delta Q/\Delta Q$) P6 fibroblasts, as evidenced by the similar staining for nuclear p53 in both cultures (Fig. 7D). Although levels of activated p53 can indicate senescence, cells can also undergo p53-independent p21 activation (43).

p16 is also a cyclin-dependent kinase inhibitor that interacts with cdk4-associated complexes. Inhibition of the cdk4

complex kinase activity by p16 results in hypophosphorylation (activation) of the retinoblastoma protein and cell cycle arrest (45). As seen with p21 expression, p16 immunofluorescence staining is also increased in the nuclei of the *Hdh*($\Delta Q/\Delta Q$) fibroblasts in comparison with wild-type controls (Fig. 7E).

To investigate further the phenotype of the *Hdh*($\Delta Q/\Delta Q$) fibroblasts, immunofluorescence staining for caveolin-1 and HIP-1 was also performed on P4 cultures (Fig. 7F). Both

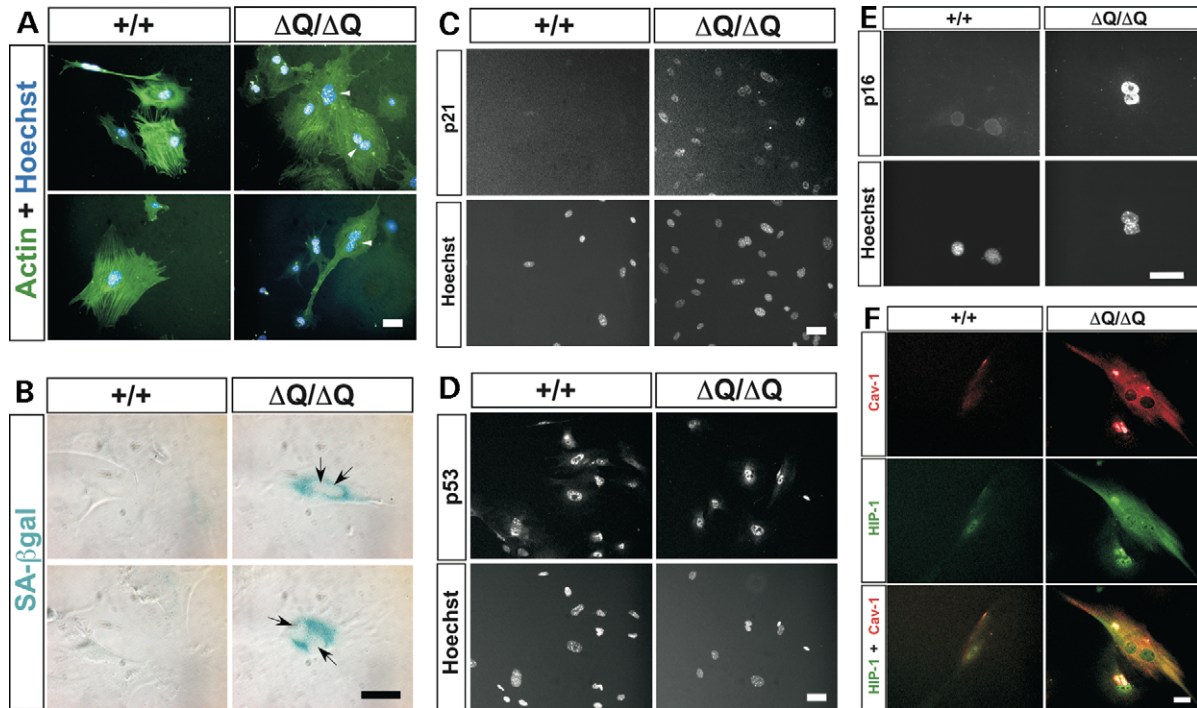


Figure 7. Enhanced expression of senescence markers in *Hdh*($\Delta Q/\Delta Q$) PMEFs. (A) Actin immunostaining (green) in control (+/+) and *Hdh*($\Delta Q/\Delta Q$) P6 PMEFs. The flattened morphology of the *Hdh*($\Delta Q/\Delta Q$) fibroblasts, many of which have double nuclei (white arrowheads), is evident. (B) Example of histochemical staining for SA- β gal activity (blue) in control (+/+) and *Hdh*($\Delta Q/\Delta Q$) P6 PMEFs. Arrows mark double nuclei that were prevalent in the *Hdh*($\Delta Q/\Delta Q$) fibroblast cultures. (C) Immunofluorescence staining for p21 in control (+/+) and *Hdh*($\Delta Q/\Delta Q$) PMEFs at P6 where elevated p21 staining in the nuclei of the *Hdh*($\Delta Q/\Delta Q$) fibroblasts is observed. (D) Immunofluorescence staining for p53 in control (+/+) and *Hdh*($\Delta Q/\Delta Q$) PMEFs at P6. In contrast to p21 staining, p53 nuclear staining was not qualitatively different in the control and *Hdh*($\Delta Q/\Delta Q$) fibroblasts. (E) Immunofluorescence staining for p16 in control (+/+) and *Hdh*($\Delta Q/\Delta Q$) PMEFs at P4. Nuclear p16 staining is elevated in the *Hdh*($\Delta Q/\Delta Q$) fibroblasts in comparison to controls. (F) Example of dual immunofluorescence staining for caveolin-1 (cav-1, red) and HIP-1 (green) in control (+/+) and *Hdh*($\Delta Q/\Delta Q$) P4 PMEFs. Elevated staining for both caveolin-1 and HIP-1 is observed in the *Hdh*($\Delta Q/\Delta Q$) fibroblasts. All scale bars = 50 μ m. Nuclei were counterstained with Hoechst in (A, C–E) (blue).

caveolin-1 and HIP-1 are htt-interacting proteins, and the expression levels of both proteins increases during senescence in culture. Although the role of caveolin-1 in senescence is still unclear, caveolin-1 expression has been observed to increase in senescent human mesenchymal stem cells (46) and murine fibroblasts (47). In addition to its role as a regulator of the androgen receptor, HIP-1 is also involved in the intrinsic apoptotic cell death pathway (30,48). HIP-1 expression is increased in cells derived from progeria patients (a human model of accelerated aging) and is also increased in fibroblasts obtained from aged normal donors (49). As with caveolin-1, the mechanism underlying HIP-1's role in senescence is still unknown. In P4 wild-type primary embryonic fibroblasts, caveolin-1 and HIP-1 immunostaining is weak, but is elevated in the *Hdh*($\Delta Q/\Delta Q$) fibroblasts (Fig. 7F). Thus, with the exception of p53 levels, the expression of four senescence-associated proteins are increased in *Hdh*($\Delta Q/\Delta Q$) fibroblasts.

To assess whether premature senescence occurs *in vivo*, brain sections from 1-year-old control and *Hdh*($\Delta Q/\Delta Q$) mice ($n = 2$) were examined for the presence of lipofuscin. Lipofuscin pigment accumulates in the cytoplasm of neurons as they age, and lipofuscin-derived autofluorescence has been found to increase with age in the brains of both mice and humans (50,51). In all brain areas examined (thalamus, cortex and hippocampus), a marked increase in the amount of lipofuscin

autofluorescence was detected in the *Hdh*($\Delta Q/\Delta Q$) brains (Supplementary Material, Fig. S1). Despite these results, we have not yet detected any other obvious physical manifestations of premature aging in the oldest of our *Hdh*($\Delta Q/\Delta Q$) mice.

DISCUSSION

The short polyQ stretch within mouse htt is not essential for htt's normal function during embryonic development. Deletion of the polyQ stretch also has little observable effect on the subcellular localization of ΔQ -htt, on the gross morphology of the adult mouse brain and on the association of ΔQ -htt with several htt-interacting proteins. The lack of an embryonic or more overt phenotype in the *Hdh*($\Delta Q/\Delta Q$) mice is not unexpected, as the size of htt's polyQ stretch is variable in humans (52), because *Drosophila* htt functions normally without a polyQ stretch (23), and deletion of polyQ or Q-rich regions in other proteins does not necessarily lead to complete functional inactivation.

For example, deletion of a polyQ stretch is tolerated for dimerization of transcriptional activators (53), formation of protein complexes associated with signaling cascades (54) and regulation of gene expression (53–55). The contribution of a polyQ stretch to normal function may also vary within the same protein for those polypeptides with multiple polyQ

or Q-rich domains. Deletion of the polyQ stretch within a regulatory protein of the nitrogen cycle (*nit-4*) does not affect its function, but co-deletion of a Q-rich domain abolishes its activity (56). Moreover, the DAL81 protein contains two polyQ stretches, one of which is dispensable, whereas the other diminishes DAL81 activity by 50% when deleted (57).

Although the polyQ stretch within mouse *htt* is not essential for *htt*'s normal function during development, deletion of the polyQ stretch results in more errors in the beginning trials of a learning/memory task, increased exploratory behavior in an olfaction test and subtly enhanced rotarod performance in adult mice. Both heterozygous and homozygous ΔQ -*htt* mice explore more holes in the Barnes circular maze before entering the escape tunnel on the first 2 days of testing. On subsequent days, they perform similarly to controls. We interpret this behavior to suggest that the ΔQ -*htt* mice are either slightly impaired in their ability to learn the task or, alternatively, they exhibit more exploratory behavior in a novel environment relative to controls.

The olfaction test involving locating a food treat also revealed a subtle behavioral difference in the *Hdh*($\Delta Q/\Delta Q$) mice. The *Hdh*($\Delta Q/\Delta Q$) mice exhibited an increased latency to find the buried treat, even in the presence of food deprivation as a motivational source. We interpret these results as another indication that the *Hdh*($\Delta Q/\Delta Q$) mice have either a tendency to exhibit more exploratory behavior than controls when placed in a novel environment or they might lack motivation for completing novel tasks. This behavioral phenotype could occur in the absence of hyperactivity, but perhaps is not reflected in the open-field test (which was performed with older mice at 9 months of age).

In addition to deficits in the Barnes circular maze and olfaction test, the *Hdh*($\Delta Q/\Delta Q$) mice exhibit a slight improvement in rotarod performance. This improvement cannot be attributed to strain differences, as C57BL/6J mice show much longer latencies to fall from the rotarod than 129/SvJ mice (58). In the literature, we are aware of only a few examples of improved rotarod performance that are the result of genetic mutation in the mouse. For example, in a mouse model for Down's syndrome (Ts65Dn) that has three copies of a portion of mouse chromosome 16 (corresponding to the human chromosome 21 trisomy mutation), improved rotarod performance was observed, as well as hyperactivity in the open field (59). Similar phenotypes were also exhibited in mice heterozygous for a mutation in the heregulin gene that encodes a ligand for tyrosine kinase receptor signaling pathways (60). These results were unexpected in both instances, as cerebellar problems were anticipated in both models because Ts65Dn mice have reduced cerebellar volume (61) and heregulin is important for proper cerebellar development (62). In addition, mice deficient for a protein repair methyltransferase enzyme (a mouse model for epilepsy) exhibited enhanced rotarod performance relative to controls on a single day of testing (63).

Although the *Hdh*($\Delta Q/\Delta Q$) mice appear to exhibit subtly enhanced motor coordination, their motor learning rate for the rotarod test does not differ from wild-type controls. Also, in contrast to previous observations of enhanced rotarod performance in mutant mice, hyperactivity was not observed in the *Hdh*($\Delta Q/\Delta Q$) mice when tested at 9 months

of age. In an attempt to determine whether the *Hdh*($\Delta Q/\Delta Q$) mutant's enhanced rotarod performance may be due, in part, to an altered cellular physiology, we obtained fibroblasts from embryos and adult ear tissue in order to characterize ATP levels in culture. Mutant *htt* with an expanded polyQ stretch, for example, can elicit phenotypes in somatic cells in addition to neurons. Lymphoblasts obtained from HD patients, as well as *Hdh*(Q111) striatal neurons in culture, exhibit reduced ATP levels or reduced ATP/ADP ratios *in vitro* (24). Interestingly, we observed elevated ATP levels in the *Hdh*($\Delta Q/\Delta Q$) fibroblasts, but not in the brain or skeletal muscle from 14-month-old mutant mice. Perhaps, the increase in ATP is a property unique to fibroblasts or it is possible that the elevated ATP levels observed in the mutant fibroblasts are a consequence of their response to *in vitro* culture conditions. If the latter interpretation is correct, the absence of any significant differences in ATP levels measured in the tissues of the control and mutant mice may be due to homeostatic processes in the context of the whole animal, which are able to control ATP levels more effectively in response to stress.

The elevated ATP level observed in the *Hdh*($\Delta Q/\Delta Q$) fibroblast cultures was accompanied by a reduction in cell proliferation. It is likely, based on the observed correlation between reduced embryonic fibroblast proliferation and elevated ATP levels, that the increase in cellular ATP is related directly to reduced growth in culture. Both p21 (an inhibitor of cdk2/cyclin E complex activity) and p16 (an inhibitor of cdk4/6/cyclin D1 complex activity) were elevated in the mutant fibroblasts, along with HIP-1, caveolin-1 and SA- β gal expression, suggesting that the reduced proliferation likely represents senescence. These results are in contrast to those obtained with skin fibroblast cultures obtained from 12-week-old R6/2 transgenic HD model mice. The R6/2 fibroblast cultures exhibited a reduced mitotic index and disorganized centrosomes, therefore making it difficult to maintain the cells in culture, but there was no evidence of premature senescence (64). These phenotypes in the R6/2 fibroblasts obtained from the 12-week-old mice were not as apparent as in the fibroblasts obtained from younger animals, suggesting that they reflected a progressive change in the skin of the R6/2 mice. In our model, we were able to obtain mutant fibroblasts from adult mice, although it is noteworthy that ATP levels were already elevated at P1 in the adult mutant fibroblasts. Further studies are needed to determine whether this represents a progressive phenotype in our *Hdh*($\Delta Q/\Delta Q$) mice.

Our observation that the *Hdh*($\Delta Q/\Delta Q$) fibroblasts were senescing prematurely and had higher levels of ATP was somewhat surprising, because there is a large body of data correlating cellular senescence with lower levels of ATP (65). However, the results from recent investigations using antagonists of the different cdk/cyclin complexes may resolve this apparent discrepancy. Roscovitine, an inhibitor of the cdk2 complex causes senescence in cell culture that is not accompanied by any change in ATP concentration, but flavopiridol, an inhibitor of the cdk4 complex, increases cellular ATP content in Hep G2 cells (66). Similarly, in leukemia cells, cell cycle arrest can be uncoupled from energy status by exogenous expression of p16, resulting in elevated levels of ATP (67). In both of these examples, experiments were performed with immortalized cells, and the mechanism linking

enhanced p16 expression and/or inhibition of cdk4 complexes to elevated ATP levels is unknown.

On the basis of the concept of the 'Hayflick limit,' it has been proposed that a correlation exists between the proliferative life span of *in vitro* fibroblasts and the longevity of the donor animal (68). Although we have detected increased lipofuscin pigment in the brains of the *Hdh*($\Delta Q/\Delta Q$) mice, we have not yet detected any other obvious signs of aging in the oldest of these mice (now 14 months old), and further observations are needed in order to determine whether additional signs of premature senescence occurs *in vivo*. Nonetheless, the elevation in ATP levels exhibited by the embryonic fibroblasts that occurs in parallel with cellular senescence suggests that the polyQ stretch contributes to a normal htt function involved in modulating longevity of primary cells in culture and, potentially, energy status. The *Hdh*($\Delta Q/\Delta Q$) fibroblasts also provide another data point to compare with the energy level status of lymphoblastoid cells from HD patients exhibiting ATP/ADP ratios that correlate inversely with htt polyQ length. In our *Hdh*($\Delta Q/\Delta Q$) primary cell cultures, energy status appears to be uncoupled from cell proliferation with high levels of ATP correlating with cellular senescence.

It is important to note, however, that the polyQ deletion results in only a subtle phenotype *in vivo*, and thus it is likely that the polyQ stretch is not required for an essential function of htt, but instead, may modulate a normal function of htt. Deletion of the polyQ stretch, for example, may influence htt's ability to act as a scaffold for diverse signaling pathways in the cell, thus impacting both cell cycle progression and energy status. In vertebrates, a short polyQ stretch in htt may be required for optimal normal function. Expansion of the polyQ stretch beyond this short optimum length could then result in altered htt function leading to the dominant energy phenotype observed in lymphoblastoid cells from both normal and HD patients, whereas deletion of the polyQ stretch results in a different set of phenotypes that are characterized by premature senescence *in vitro* and subtle behavioral abnormalities *in vivo*.

MATERIALS AND METHODS

Generation of *Hdh*($\Delta Q/+$) mice

Introduction of the ΔQ mutation and N-terminal FLAG epitope modifications of *Hdh* exon 1 were performed in two sequential steps by replacing first an endogenous *Hdh* exon 1 *XmnI*–*NarI* restriction fragment containing the polyQ stretch of the htt coding sequence, with a synthetic *XmnI*–*NarI* fragment that was generated by annealing two complementary oligonucleotides carrying the ΔQ mutation (deltaQ-1: *XmnI* site double underlined, *NarI* site single underline; 5'-GATGAAGGCTTTCGAGTCGCTCAAGTCGTTT><CCACCGCCGCAGGCGCCGCCG-3', deltaQ-2: 5'-CGGCGGCAGCTGCGGCGGTGG><AAACGACTTGAGCGACTCGAAGCCTTCATC-3'). Following annealing, the DNA was digested with *XmnI* and *NarI* and then cloned into an *XmnI*–*NarI* vector prepared from a 1.6 kb *PstI* *Hdh* genomic fragment subclone containing the *Hdh* promoter, exon 1, and a portion of intron 1. To introduce the FLAG epitope modification, an

AlwNI–*XmnI* fragment containing the N-terminus of htt was replaced with a synthetic *AlwNI*–*XmnI* fragment that was generated by annealing two partially complementary oligonucleotides containing the FLAG epitope-modified N-terminal sequence and appropriate restriction enzyme sites (FLAG forward: *AlwNI* underlined, complementary bases in italic, 5'-GTCTTCAGGGTCTGTCCCATCGGGCAGGAAGCCGTCATGGACTACAAGGACGACGATGACAAG-3', FLAG reverse: *XmnI* double underlined, 5'-CGACTCGAAAGCCTTCATCAGCTTTTCCAGGGTTGCCTTGTCATCGTCGTCCTTGTA GTC-3'), and then extending the annealed partial duplex with Klenow fragment of DNA Polymerase I enzyme and deoxynucleotide triphosphates. The synthetic DNA fragment was then digested with *AlwNI* and *XmnI* restriction enzymes and cloned into the ΔQ -modified 1.6 kb *PstI* genomic fragment. The ΔQ and FLAG modifications were verified by DNA sequencing, and a gene targeting vector was assembled using 7.1 kb of 5'-flanking homology, a floxed *pgkneo* cassette inserted within a *HindIII* restriction site located 1.3 kb upstream of exon 1 for positive selection of transfected ES cells and ~2.8 kb of 3'-flanking homology. The gene-targeting vector was linearized at a *NotI* site located adjacent to the 5'-homology region and electroporated into W9.5 ES cells (129Sv background). ES clones were screened by Southern blotting of *NcoI* digested genomic DNA with a 3'-flanking probe located outside the targeting vector 3'-homology (150 bp *XhoI*–*HindIII* fragment) (7). The 3'-flank probe recognizes a 10 kb wild-type fragment and a 12 kb fragment from the targeted allele (2 kb larger in size because of the addition of the floxed *pgkneo* cassette). To determine whether the targeted ES clones identified by Southern analysis also contained the FLAG and ΔQ modifications, PCR amplification of the epitope tag and ΔQ regions of exon 1 was performed. For amplification of sequence containing the ΔQ mutation, the 5' (forward) oligonucleotide is located near the 5' end of exon 1, while the 3' (reverse) oligonucleotide spans the CAG repeat deletion site, and is thus able to anneal to the mutant, but not to the wild-type allele, to generate a 226 bp product. ΔQ forward: 5'-GACGGGCCCCAAGATGG-3'; ΔQ reverse: 5'-GGCGGTGGAAACGACTT-3'. Cycle conditions: 3 min 30 s initial denaturation at 94°C, followed by 30 cycles consisting of 20 s at 94°C, 30 s at 60.8°C and 55 s at 72°C with a 3 s/cycle elongation, followed by a final 5 min extension at 72°C. Although the ΔQ PCR product also contains the FLAG sequence, for simultaneous detection of both the epitope tag and the wild-type sequence, oligonucleotide primers flanking the site of the FLAG insertion were used to generate either a 112 bp product (wild-type allele) or a 136 bp product (targeted allele). The 5'-oligonucleotide is located 27 bp upstream of the *AlwNI* restriction site, whereas the 3'-oligonucleotide is located just 5' of the ΔQ deletion site. Epi-forward: 5'-GCGTAGTGCCAGTAGGCTCCAAG-3'; Epi-reverse: 5'-CTGAAACGACTTGAGCGACTCGAAAG-3'. Cycle conditions: 2 min initial denaturation at 94°C, followed by 35 cycles consisting of 20 s at 94°C, 20 s at 70°C and 20 s at 72°C with a 1 s/cycle elongation, followed by a final 5 min extension at 72°C. Targeted ES clones were obtained with a frequency of 17%, and two ES clones ($\Delta Q57$ and $\Delta Q62$) were selected at random and the exon 1 sequence of their targeted alleles was amplified by PCR and sequenced for verification: forward oligonucleotide = Epi-forward and reverse oligonucleotide =

Intron 2; 5'-GAGACCCCGCAAGACGAGGG-3' located ~150 bp downstream of the end of exon 1. The two ES clones were then injected into C57BL6 blastocysts and germline chimeras were obtained using standard procedures (69). All protocols for animal use were approved by the Institutional Animal Care and Use Committee of the University of Virginia, and were in accordance with the NIH guidelines. The presence of the *Hdh*^{ΔQ} allele in progeny of the germline transmitters was confirmed using Southern analysis, and sequence was obtained from the exon 1 region of the targeted allele using tail DNA isolated from *Hdh*(ΔQ/+) pups to ensure that the epitope tag and ΔQ mutation was present and that there were no other mutations in the region (Epi-forward and Intron 2 oligonucleotides). For routine genotyping, PCR is used to confirm the presence of the ΔQ mutation and additional PCR analyses are performed to detect the epitope tag within exon 1 (discussed earlier) and the *pgkneo* cassette upstream of the transcription start site (pgk-1: 5'-GCCCCGGCATTCTGCACGCTT-3', neo-2: 5'-GAGTACGTGCTCGCTCGATG-3', 534 bp product; cycle conditions are: 1 min 30 s initial denaturation at 94°C, followed by 35 cycles consisting of 20 s at 94°C, 30 s at 65°C and 30 s at 72°C with a 1 s/cycle elongation, followed by a final 5 min extension at 72°C).

Motor and behavioral analyses

All tests used groups of homozygous mutants and wild-type controls, with the exception of the Barnes circular maze, where a separate group of heterozygous mutants was also tested.

Rotarod. The mice used for the rotarod test were derived from 10 different litters. One-third of the mice tested were derived from ES clone 62, whereas the remaining mice were derived from ES clone 57. Testing included 2 or 3 separate initial trial days (depending on the time point) to control for environmental parameters, and a mix of genotypes were tested on each day. Animals were tested at 1 month ($n = 17$ for each genotype), 4 months ($n = 15$ for each genotype) and 9 months ($n = 13$ for each genotype) of age on an Economex accelerating rotarod (Columbus Instruments) that has the capacity to test four mice simultaneously. The testing procedure consists of three phases: stationary training on the non-rotating rod, training on a rod that is rotating at a constant 2.5 r.p.m. and five consecutive days of testing on an accelerating rod.

Stationary training. The mouse is placed on the stationary rod facing the back wall of the apparatus, and the mouse is allowed to remain on the rod until it falls to the bottom of the chamber or 60 s have elapsed. This procedure is repeated two more times (for a total of three trials) with no inter-trial interval. If the mouse has stayed on the rod for a total of 60 s over the three trials, then it proceeds to training on the rod rotating at a constant velocity.

Training on the rotating rod. The rotarod is adjusted to 2.5 r.p.m. and the mouse is then placed on the moving rod, facing the back wall of the apparatus. The mouse is allowed

to stay on the rod until it falls or 60 s have elapsed. This procedure is repeated two more times (for a total of three trials) with no inter-trial interval. If the mouse remains on the rod for a total of 60 s over the three trials, then it moves on to the accelerating rod test.

Accelerating rotarod test. The rotarod is adjusted to 2.5 r.p.m. and the mouse is placed on the rod facing the back of the apparatus. If the mouse falls from the rod while other mice are placed in the remaining three compartments, the mouse is put back onto the rod. If the mouse falls again, it is placed back on the rod one more time. If the mouse fails again to remain on the rotating rod, it is given a score of 0 s. Once all the mice have been placed on the rotating rod, the acceleration is started (6 r.p.m.; the rod is rotating at ~32.5 r.p.m. after 300 s) and the latency to fall is recorded. The accelerating test is repeated two additional times with an average inter-trial interval of 10 min (total of three trials/day). The accelerating test is performed each day for a total of five consecutive days. Significance was assessed using repeated measures two-way ANOVA, Bonferroni *post hoc* test.

Barnes maze. The mice used for the Barnes maze were derived from eight different litters. Approximately one-third of the mice tested were derived from ES clone 62, whereas two-thirds were derived from ES clone 57. Testing included three separate initial trial days over a period of 6 months to control for environmental parameters, and a mix of genotypes were tested on each day.

Set up. At 5 months of age, the animals were tested once a day during daytime hours for 9 days ($n = 6$ for each genotype). The Barnes maze (San Diego Instruments) was placed in the center of an isolated 15 × 10 ft.² room, with geometric shapes arranged on the wall for visual orientation purposes. The maze consisted of a round center platform ($D = 112.5$ cm) elevated 3 ft. off the floor. The perimeter of the circular platform was ringed with 20 holes ($D = 5$ cm), one of which led to an escape tunnel (20 × 9 cm²) hidden from view under the platform. The experimenter remained hidden behind a curtain during the testing.

Procedure. Each animal was placed individually in a center holding chamber (15 × 20 cm²) oriented in an arbitrary position. First, a buzzer located under the platform was turned on, followed by a 125 W fluorescent bulb located 45 cm above the center of the platform. After 10 s, the center chamber was lifted and the animal was allowed to explore the maze. The first trial was a shaping trial, where the animal was guided directly to the escape tunnel. Once the animal entered the tunnel, the light and the buzzer were turned off and it was allowed to remain in the tunnel for 2 min while the surface of the maze was cleaned with Spor-Cleanz and 70% ethanol. After habituation, the animal was removed from the tunnel and placed in the center holding chamber once again for 10 s. Trial one followed immediately.

The tunnel was placed under a randomly selected hole for each animal, but the tunnel was located in the same position for each trial. The order in which the animals were tested varied each day. To control for odor cues, the platform surface was wiped down with both Spor-Cleanz and 70% ethanol between animals and the entire platform was rotated 90° each day. Animals were tested for 9 days.

During initial trials, the mice were attempting to sit in the shallow non-target holes, although they could not enter them completely. The holes were lined with tissue (Kimwipes) to make them smaller. It was also noticed that the animals did not readily enter the tunnel hole once they located it, so a hidden ($9 \times 3 \times 3 \text{ cm}^3$) step was added to make the tunnel easier to enter.

Scoring. A nose-poke into a hole was scored as an error. All four paws had to be inside the tunnel in order for the trial to end. Once the animal entered the tunnel, it was allowed to remain in the tunnel for 2 min. If the animal did not enter the tunnel within 5 min, it was placed there and allowed to remain for 2 min. Each animal was given a distance score that reflects how far the first hole visited was away from the target hole (e.g. a hole $\sim 160^\circ$ from the escape hole was given a numerical value of 9; one hole closer on either side, a value of 8) (Fig 4A). In addition, the latency to find the target, latency to enter the target, number of errors before finding the target and total number of errors were recorded. Significance was assessed using repeated measures two-way ANOVA, Bonferroni *post hoc* test.

Olfaction tests. **Digging test.** Olfaction was assessed by a paradigm requiring animals to find food buried in the cage bedding at 7 months of age ($n = 5$ for each genotype). Initially, group-housed female mice were exposed to a treat (1 cm^3 chunk of Nutri-Grain bar) once daily for 3 days in the home cage. On the fourth day, the treat was placed in the back left corner of a new cage ($20 \times 30 \text{ cm}^2$). Approximately 2.5 cm thick layer of clean bedding was placed in the cage uniformly to cover the location of the treat. Each animal was placed individually in the front right corner of the cage. Latency to dig anywhere in the bedding and latency to find the treat was recorded (up to 300 s). The mouse had to touch the buried treat for the timer to stop. At 8 months of age, the same olfaction test was repeated with a food deprivation of 36 h and a chocolate treat to ensure the animals were properly motivated to find the food.

Bedding test. At 7 months of age, female mice were individually placed in a 38 cm L $\times$ 20 cm W $\times$ 10 cm H plexiglass box with a wire floor elevated 2.5 cm off the bottom of the cage. A shallow tray that contained bedding from the mouse's home cage on one side and clean bedding on the other side was placed beneath the elevated wire floor. The mouse was placed in the middle of the wire floor and allowed to explore for 3 min. The amount of time all four paws were located over the home bedding was recorded.

Elevated plus maze. Behavior in the elevated plus maze was assessed at 3 months of age ($n = 10$ for each genotype). The maze had equally sized (35 cm long $\times$ 6 cm wide) open

arms and closed arms. The two closed arms had 20 cm high clear plexiglass walls. The maze was elevated 50 cm above the floor. The animals were tested individually in a $10 \times 15 \text{ ft}^2$ room. At the start of the test, a mouse was placed in the center of the maze facing a closed arm and was allowed to freely explore the maze for 5 min. The total number of arm entries, the proportion of open arm entries and the proportion of time spent in the open arms were recorded. The test was videotaped for later scoring. Significance was assessed using the unpaired Student's *t*-test. In this test, neither the time spent in open arms nor the number of arm entries differed significantly between controls and mutants.

Light/dark box. The test was performed at 6 months of age during daytime hours ($n = 9$ for each genotype). The same cohort of mice tested in the elevated plus maze was used for the light/dark box test. The light/dark box consisted of a normal housing cage divided into light and dark compartments. These compartments were connected by a door ($4 \times 4.5 \text{ cm}^2$) located in the center of the partition at floor level. The light section ($18.5 \times 19 \text{ cm}^2$) was open at the top and painted white. The dark box ($18.5 \times 10 \text{ cm}^2$) was painted black and had a removable black lid. The mouse was placed in a corner of the light box from the top. The latency to enter the dark box (defined as placement of all four paws inside the dark box), the number of entries and the amount of time spent in the dark box during a period of 5 min were measured manually. The test was videotaped for later scoring. All differences between controls and mutants were found to be not significant using the unpaired Student's *t*-test.

Open-field testing and activity monitoring. Behavior in an open field was also measured using a Versamax activity monitor (AccuScan Instruments, Columbus, OH, USA) for mice at 9 months of age ($n = 8$ for each genotype). Each mouse was placed initially in the bottom left corner of the monitoring cage, and the time spent in the center area of the field was compared with the time spent within two bodies' width of the walls over a 5 min interval (test period between 1 and 4 p.m.). The data were analyzed using Versamax software. In addition, animal activity levels (horizontal movements) were examined during the open-field test using Versamax software. Significance was assessed using Student's *t*-test.

Histological analyses

Mice were euthanized by cervical dislocation following administration of isoflurane. The brains were removed and frozen rapidly in methyl-butane on dry ice. Brains were sectioned at 30 μm on a cryostat (Bright Instrument Co.) and stored frozen at -80°C until use.

Cresyl violet staining. Sections were thawed and a Cresyl violet stain was performed according to manufacturer's protocol (FD Cresyl Violet Solution, FD Neurotechnologies, Ellicott City, MD, USA) ($n = 3$ for each genotype).

Subcellular fractionation

PMEFs were isolated from E13.5 embryos according to standard procedures (70). Confluent fibroblasts from a 10 cm tissue culture dish were suspended in 0.2 ml cell lysis buffer (5 mM PIPES pH 8, 85 mM KCl, 0.5% NP-40) containing a protease inhibitor cocktail (Complete Mini, EDTA-free; Roche Diagnostics) and incubated on ice for 30 min. Crude nuclei were pelleted with a 4 min spin at 8200g, 4°C and after washing with 500 µl phosphate-buffered saline (PBS), the crude nuclear pellet was resuspended by dounce-homogenization in 300 µl nuclear lysis buffer [1% sodium dodecyl sulfate (SDS), 10 mM EDTA, 50 mM Tris-HCl pH 8.0 plus protease inhibitors] and incubated 30 min on ice. A sonicator was used (4 × 2 s bursts) to shear DNA, and protein concentration was determined using a bichinonic acid assay (Pierce).

Western analysis

Samples were fractionated on SDS-PAGE and transferred onto a PVDF membrane (Invitrogen) at 30 V overnight at 4°C. The membrane was rinsed in methanol 1 min, water for 3 min and TBST [1 × Tris-buffered saline (TBS), 0.05% Tween-20] for 5 min. The membrane was then blocked (5% non-fat dry milk, 2% heat-inactivated goat serum, 1 × TBS, 0.05% Tween-20) 1 h at room temperature, washed twice with TBST briefly, then 1 × 5 min and then incubated with primary antibody solution overnight [MAb 2166 (Chemicon) at 1:4000, HIP-1 (Santa Cruz) at 1:500, p53 (NovoCastra) at 1:1000, cytoplasmic marker Grb2 (Santa Cruz) at 1:1000] diluted in a 1:10 block:TBST solution shaking at 4°C. The membrane was washed twice with TBST briefly, 1 × 15 min, then 4 × 5 min, before incubation with a goat anti-mouse IgG or goat anti-rabbit IgG conjugated to horse radish peroxidase secondary antibody (Jackson Immunologicals); 1:50 000 diluted in a 1:10 block:TBST solution for 1 h at room temperature. The blot was then washed as described earlier, incubated 5 min in chemiluminescence substrate (Supersignal Femto West, Pierce) and then exposed to film. For densitometry, films in the linear exposure range were scanned on a flatbed scanner (Canonscan D2400U, Canon) and analyzed using the ImageJ program (Rasband, W.S., ImageJ, U.S. National Institutes of Health, Bethesda, MD, USA, <http://rsb.info.nih.gov/ij/>, 1997–2005). Htt levels were corrected for loading variation by normalizing to the amount of GAPDH in each sample.

Co-immunoprecipitation

Whole brains were isolated from mice (10 months old) and dounce-homogenized in 1 ml of lysis buffer (Subcellular Fractionation). Co-immunoprecipitation was performed using the Catch and Release v2.0 Reversible Immunoprecipitation System (Upstate), using 1 mg of protein extract and 3 µl of MAb 2166 (Chemicon) according to manufacturer's protocols. Western antibodies used were FLAG M2 (Sigma) at 1:2500, 2166 (Chemicon) at 1:4000, HIP-1 (Santa Cruz) at 1:500, PSD-95 (BD Transduction Laboratories) at 1:1000, GAPDH (Chemicon) at 1:1000.

Immunohistochemistry

Fibroblasts were grown to 60% confluency in slide chambers (Nunc). Slides were fixed with 4% paraformaldehyde in PBS for 10 min at room temperature, followed by incubation with a 1:1 solution of 4% paraformaldehyde and 0.1% Triton X-100 in PBS for 10 min at 37°C. Slides were washed once in 0.1% Triton X-100 in PBS and incubated in 0.2 N HCl in water for 10 min at room temperature, followed by washing twice with PBS. The slides were then washed in 5 mM MgCl₂, 0.1% Triton X-100 in PBS for 5 min at room temperature before being washed twice with PBS. Cells were blocked for 30 min at room temperature in a solution containing 10% goat serum and 0.1% Triton X-100 in PBS and then exposed overnight at 4°C to the primary antibody MAb 2166 (Chemicon), p53 (NovoCastra), p21 (Santa Cruz), p16 (Santa Cruz F-12), actin (MP Biomedicals), HIP-1 (Santa Cruz), caveolin-1 (BD Transduction Laboratories), diluted in the blocking reagent at 1:100. The next day, cells were washed thrice with 0.05% Triton X-100 in PBS and then incubated 1 h in the dark with the secondary antibody (donkey anti-mouse or donkey anti-rabbit-Cy3 or -FITC 1:700; Jackson Immunologicals) diluted in blocking solution. After washing thrice with 0.05% Triton X-100 in PBS, the fluorescent DNA stain Hoechst 33342 was added to each slide and incubated at room temperature in the dark for 5 min. Finally, cells were washed thrice in 0.05% Triton X-100 in PBS and cover slipped using Vectashield. Slides were examined under epifluorescence using an Olympus BX51 microscope equipped with a Magnafire CCD camera. Immunohistochemistry was also performed on fresh-frozen brain tissue (25 µm sections) according to the same protocol (*n* = 2 for each genotype). All images comparing immunofluorescent staining in wild-type and mutant cells and brain sections were taken with identical exposure parameters.

SA-β-galactosidase staining

PMEFs (P6) were grown in chamber slides to ~50% confluency. Slides were washed once with PBS and fixed for 30 min at room temperature using 3.2% glutaraldehyde and 2% formalin in PBS. After fixation, slides were washed twice in PBS and then incubated in X-gal staining solution (1 mg/ml X-gal, 40 mM citric acid/sodium phosphate pH 6.0, 5 mM potassium ferrocyanide, 5 mM potassium ferricyanide, 150 mM NaCl, 2 mM MgCl₂) overnight at room temperature. The next day, the slides were washed with PBS twice and mounted using 50% glycerol/PBS. The average number of cells positive for SA-βgal staining were obtained from eight fields for each genotype.

Fibroblast growth and ATP assays

PMEFs were isolated from E13.5 embryos and plated into individual 10 cm dishes. Cells were grown until 80% confluency and then frozen until genotypes were confirmed by PCR. Once genotypes were obtained, cells from three embryos of each genotype were thawed, pooled together and then plated onto a 10 cm dish. Cells were grown until the fastest growing cultures reached 80% confluency, trypsinized

and then cell counts were performed (an average of four counts using a hemacytometer). Each trypsinization was counted as a passage. For each passage, cells were re-plated at a density of 3×10^5 cells/10 cm dish, and cell counts were obtained again once the fastest growing cultures had reached 80% confluency. After each cell count, medium containing 1000 cells was removed ($n = 4$), and the volume was adjusted to 100 μ l using fresh medium. First, 100 μ l of 0.6% trichloroacetic acid was added, and the samples were incubated 30 min at 4°C. Then, 400 μ l 250 mM Tris-acetate pH 7.75 was added, and 100 μ l aliquots (in duplicate) were removed from each sample for ATP analysis. ATP levels were measured using the Enliten ATP Assay System Bioluminescence Detection Kit (Promega). Each sample was mixed with 100 μ l rL/L reagent, vortexed briefly and light output was then measured immediately using a luminometer (TD-20/20, Turner Designs). Results were compared with an ATP standard curve to calculate moles of ATP/cell. To isolate fibroblasts from adult mice, ear biopsies were performed on 13-month-old male wild-type and *Hdh*(Δ Q/ Δ Q) mice ($n = 1$ of each genotype). Animals were anesthetized briefly with isoflurane before 3 mm biopsies were obtained from the outer ear flap. Tissue was placed in 0.25% trypsin–EDTA for 10 min, chopped into fine pieces using scissors and then cells were plated in one well of a 12-well dish containing Dulbecco's modified Eagle's medium (DMEM) containing 20% fetal bovine serum, 1% pen/strep mixture and 1% L-glutamine. ATP measurements were taken at P1 as described earlier. For ATP measurements in tissue, frontal cortex and skeletal muscle samples were obtained from 14-month-old male control ($n = 3$) and homozygous mutant mice ($n = 4$). Tissue samples were weighed and then dounce-homogenized in DMEM. Aliquots of the homogenate were then used to measure ATP levels as described earlier.

Lipofuscin analysis

Fresh–frozen brain sections (25 μ m) were thawed from storage at -80°C , washed once with PBS and then fixed for 15 min in 4% paraformaldehyde in 0.1 M PB. Slides were then washed twice with PBS, mounted with Vectashield, cover slipped and then examined under epifluorescence illumination. Autofluorescent lipofuscin deposits were visible at all wavelengths examined, and images comparing autofluorescence in the control and *Hdh*(Δ Q/ Δ Q) sections ($n = 2$ for both genotypes) were taken with identical exposure parameters.

Statistics

Data were analyzed using the paired and unpaired Student's *t*-test, ANOVA and two-way repeated measures ANOVA. Error bars represent SEM unless otherwise stated. All statistical analyses were performed using the SigmaStat program (Systat software). Significance was accepted at $P \leq 0.05$.

SUPPLEMENTARY MATERIAL

Supplementary Material is available at HMG Online.

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